



Jackie Lee

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TECHNICAL SKILLS

Programming Languages: C++, Python, C, MATLAB, Simulink, Julia, PyTorch

Tools/Frameworks: ROS2, Isaac Sim, Gurobi, Linux, Numerical Algorithms, Statistical Methods, CATIA, OpenCV, PCL

Core Expertise: SLAM, Autonomous Systems, Closed-loop Control, Sim-to-Real Transfer, Vehicle Dynamics, HIL

Hands-on Skills: Mobile Manipulator Integration, LiDAR/Camera/IMU Fusion, Mechanical Prototyping, Soldering

EDUCATION

Texas A&M University

College Station, TX

Ph.D. in Mechanical Engineering, Autonomous Systems Lab, GPA: 3.8/4.0

8/2022- exp. 5/2027

- **Research Focus:** Scalable optimization algorithms for multi-agent systems, distributed decision-making, simulation-based performance analysis. (Advisor: Dr. Sivakumar Rathinam)

Kyungwoon University

South Korea

B.S. in Unmanned Aerial and Autonomous Vehicle Engineering, GPA: 3.9/4.0

3/2018 – 2/2022

- **Awards:** Summa Cum Laude. Full-tuition Scholarship for four years; National Science and Engineering Scholarship.

EXPERIENCE

Magna Electronics Inc., ADAS Engineering

Lowell, MA

ADAS Engineering Intern (Advisor: Ashesh Goswami)

5/2026–8/2026

- Calibrated the perception front end of an automated **valet parking** system by computing **fish-eye camera intrinsics** and **camera-IMU extrinsics** to anchor downstream localization, free-space estimation, and closed-loop parking control.
- Brought up **GMSL2/FAKRA automotive cameras on NVIDIA Jetson AGX Orin**, modifying and rebuilding the vendor camera driver for the team's target JetPack/L4T configuration, and authored the bring-up documentation the team now works from.
- Containerized the perception development environment in **Docker with ROS2** to give the team a reproducible, hardware-in-the-loop build shared across developers.
- Developed collaboratively through **Git pull requests and peer code review** in a multi-developer production codebase, and validated results on bench and in-vehicle setups rather than in simulation alone.

Amazon Robotics, Warehouse Mobile Robot R&D Team

North Reading, MA

Applied Scientist II Intern (Advisor: Dr. Yuri Ivanov)

5/2025–12/2025

- Built a physics-consistent ROS2 + Isaac Sim + Unreal Engine **automated test framework** for large-scale evaluation of **motion control, PID controller performance, navigation, and perception** pipelines under sensor noise, dynamic disturbances, and contact-rich warehouse environments.
- Designed and validated **position and torque control** pipelines with EKF-based state estimation, integrating **LiDAR-based** 3D detection with real-time trajectory execution; performed **system identification** and **sim-to-real transfer** onto embedded platforms (Raspberry Pi, CAN bus, TI board, ESP32) for hardware deployment.
- Bridged research prototypes into reusable ROS2 modules and deployable robotic workflows, establishing end-to-end **simulation-to-hardware validation** pipelines for long-term system integration and ownership.

Texas A&M University, Autonomous Systems Lab

College Station, TX

Research Assistant (Advisor: Dr. Sivakumar Rathinam)

8/2022 – present

- Built scalable simulations to support **learning-based policy evaluation** and **imitation-learning** integration.
- Designed real-time coordination and task-allocation algorithms for **mobile manipulation systems** (Clearpath UGV + Universal Robots 5), enabling perception-aware base–arm integration for constrained task execution.
- Developed optimization-driven motion strategies for heterogeneous UAV-UGV systems, achieving **10–20%** energy efficiency gains and up to **30%** improvement in obstacle-dense environments, and built **physics-aware simulation** environments integrating depth cameras, LiDAR, radio, and magnetic-field sensors to evaluate closed-loop autonomy and **perception-driven** motion under dynamic environmental constraints.
- Formulated large-scale combinatorial battery optimization frameworks (3K+ geo-coordinates) using Gurobi.

DEVCOM Army Research Lab (Advisor: Dr. Sivakumar Rathinam, Dr. James Humann)

College Station, TX

Multi-Agent Path Planning and Scheduling Problem for Surveillance System

8/2022 – 7/2024

- Devised a **Python simulation system** solving NP-hard heuristic path planning problems for **4** aerial / ground vehicles to do the **72-hour** persistent surveillance and task-solving project.
- Formulated optimization-based motion strategies for heterogeneous robotic systems with physical and battery constraints,

bridging planning and execution layers. Collaborated with UGV hardware teams to validate planning algorithms through integrated simulation and field tests.

Kyungwoon University, Autonomous and Intelligent Robotics Lab

South Korea

Independent Project: PAV tri-eVTOL, Launch type foldable drone - Project Leader

9/2020 – 11/2021

- Led a team to design a **tilt-rotor eVTOL control system** using PX4 autopilot (C++), optimizing motor efficiency through real-time convex optimization.
- Won **\$13K** in the 2021 International PAV Technology Contest for a scalable UAV simulation model.
- Created a launch-type foldable drone mortar dynamics mechanism and control system using Pixhawk C. Developed a hand-layered carbon composite drone, including its 3D model, and designed a Raspberry Pi drone with its control system using MAVproxy, enhancing structural dynamics and control efficiency.

AWARDS & HONORS

Hanwha Aerospace CEO Award, the Third National University Student Capstone Contest - Team Leader

- J. Lee, H. Nar, J. Park (Nov. 2021), “Development of Launch-type foldable quad copter”, the Society for Aerospace System Engineering, The Third National University Student Capstone Contest, Gyeongju, Korea.

Second Place, 2021 International PAV Technology Contest

- Won \$13,000; Downsized Uber-Hyundai SA-1 PAV eVTOL (using 12 motors) dynamics and control.

Third Place, 2019 Korea IT Businesswomen’s Association (IBWA) ICT Mentoring - Team Leader

- Developed a Machine Learning-based hidden camera detecting technology, app, and HTML website

PUBLICATIONS

- Lee, J.J., Rathinam, S. “Collaborative UAV-UGV Area Coverage under Battery Constraints: Team Orienteering Path Planning and Optimization-Based Scheduling.” *ICUAS*, Feb. 2025.
- Kumar, N., Lee, J.J., Rathinam, S., Darbha, S., Sujit, P.B., & Raman, R. “The Persistent Robot Charging Problem for Long-Duration Autonomy.” *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2191–2198, Mar. 2025.
- Lee, J.J., Rathinam, S. “A Meta-Heuristic Approach for an Aerial-Ground Vehicle Path Planning Problem.” *AIAA SCITECH*, Jan. 2024.

CERTIFICATION

- FAA Part 107 Certified Remote Pilot | Oct 2022
- **Ultralight Vehicle Pilot** | Korea Transportation Safety Authority | Jun 2020

LEADERSHIP EXPERIENCE

Korean-American Scientists and Engineers Association (KSEA), President and Secretary

8/ 2022 – 8/2025

- Organized the 2025 US-Korea Conference (UKC) as an American secretary for the coordinating nationwide planning teams and organizing **technical workshops**, coordinating **1500+** attendees and **200+** industry/academic speakers.
- Led the Texas A&M chapter for one year, orchestrating multi-institutional events and a career development forum featuring university faculty

Kyungwoon University Club Grin-Narae, Founder and President

3/2019 – 12/ 2020

- Established a drone development club focusing on modular solutions for land, sea, and aerial applications.
- Secured a 2020 Prospective Start-up Package grant from the Korea Institute of Startup & Entrepreneurship Development for “Non-face-to-face Emergency Rescue PAVs.”
- Won 2020 Designated University Startup Club funding from the Korea Expressway Corporation, enabling lightweight quadcopter production using DJI Inspire 2 materials.

TEACHING EXPERIENCE

MEEN 365: Dynamic Systems and Control Lab

1/2025-present

- Taught students hands-on instruction on AC/DC motors and slider-pendulum system measurement and guided them through calibration and tuning PID using MATLAB and Simulink for dynamic systems and control lab projects.
- Mentored **88+** students on HIL experimental setup, data acquisition, program troubleshooting, and effective use of MATLAB and Simulink to ensure accurate data measurement and recording for dynamic systems and control lab projects.
- Assisted **270** students by answering questions and clarifying concepts in the Dynamic Control Systems.

MEEN 225: Engineering Mechanics (honor class)

8/2024-12/2024

- Facilitated recitation sessions to reinforce lecture material and improve student understanding.
- Assisted **250+** students by answering questions and clarifying concepts in Engineering